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Telephony and SMS

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Session Agenda

- Telephony Overview
- Telephony Events
- Telephony APIs
- SMS Overview
- SMS Library

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What Is the Telephony Manager?

Telephony Manager gives you a set of APIs to control cell phones and wireless modems through the Palm device

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Telephony Goals

- Services to support wireless applications
  - Instant messaging
  - Phone book management
  - Basic voice features
- 1-piece and 2-piece configurations
- Network protocol independence
  - GSM, CDMA, TDMA, PDC…
  - However, the first version will only support GSM

Telephony Operation



End User applications: Address Book, Calendar, Memo, etc.

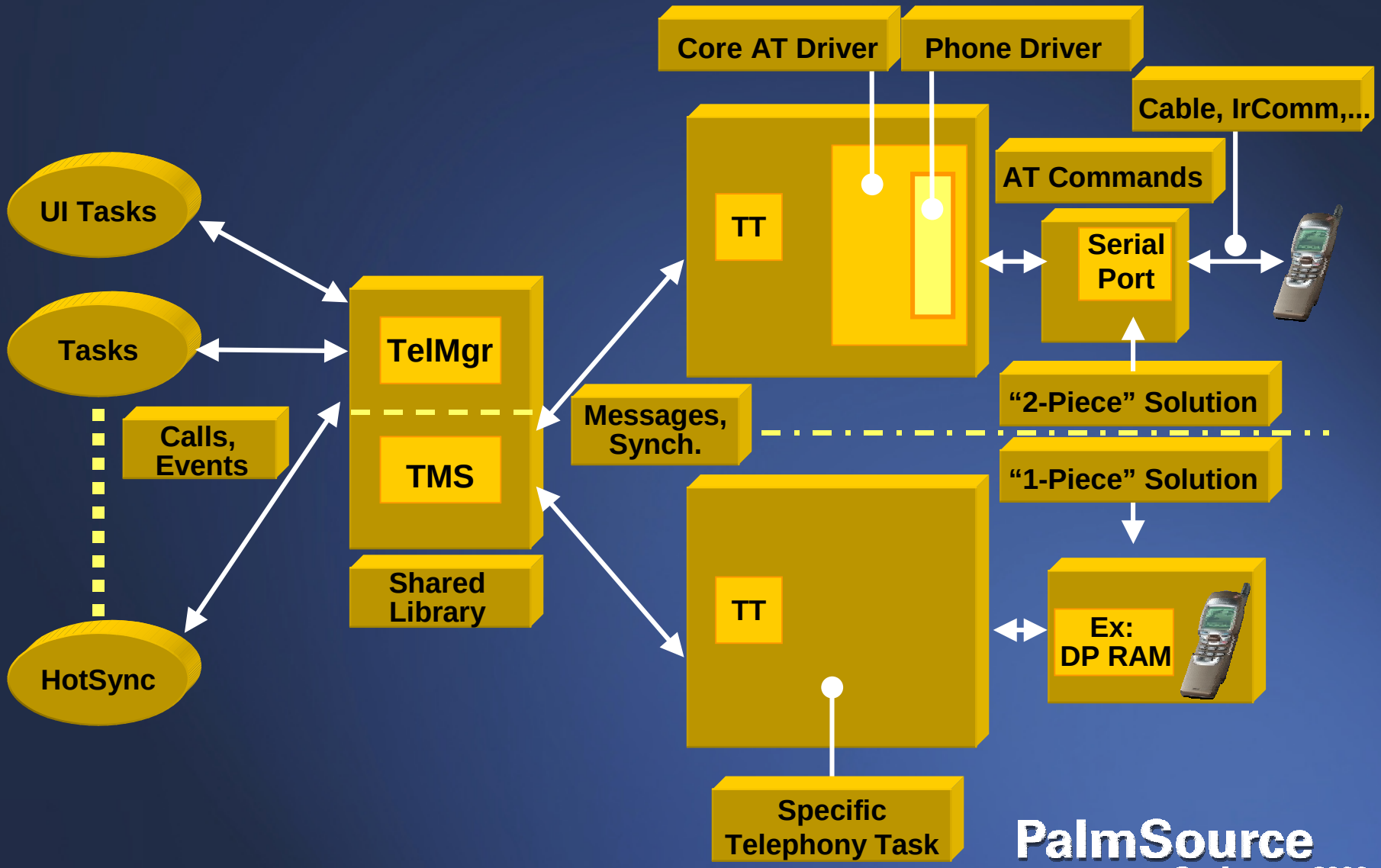
Telephony Manager (shared library) gives access to phone related features

Messaging system exchanges requests/results between TelMgr and Telephony Task

Abstracts telephony services from low-level phone hardware/software

Phone dependent driver (manufacturer)

Telephony Overview



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Telephony Integration

- Makes use of Notification Manager
  - Processes unsolicited events and sub-launches registered app or calls registered callbacks
- All Telephony components are replaceable
  - Network-specific Telephony tasks can be developed later
- Phone Manufacturer supplies phone driver
  - Many drivers can be loaded on your Palm
  - Only one can be used at a time
  - Palm supplied “standard GSM driver” is GSM 07.07 and 07.05 compliant

Telephony SDK

- Full set of APIs for accessing phone services
- All functions can be synchronous or asynchronous
  - Synchronous = Send Query and waits for response
  - Asynchronous = Sends Query and receives event code later
  - Asynchronous functions can be cancelled

Telephony Events

- TelGetEvent
  - Retrieves both telephony and standard Palm OS® events
  - Internally it calls EvtGetEvent and TelGetTelephonyEvent
- TelGetTelephonyEvent
  - Only returns telephony events

Telephony Events

- TelEventType is very similar to EventType

```
typedef struct _TelEventType
{
  eventsEnum eType;
  Boolean    penDown;
  UInt8      tapCount;
  Int16      screenX;
  Int16      screenY;
  UInt16     functionId;  /* function called */
  UInt16     transId;     /* Match call with result */
  MemPtr     *paramP;     /* parameter passed */
  Err        returnCode;  /* function return code */
} TelEventType;
```


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Telephony Service Sets

- Phone functionality differs from one phone to the next
  - Check if a specific function is supported
    - e.g., **`TelIsNwkGetNetworkNameSupported()`**
  - Check if a set of functions is supported
    - e.g., **`TelIsPhbServiceAvailable()`**

Telephone APIs (1 of 4)

- TelNwk prefix
  - Network-oriented services (current network, signal level, search mode, etc.)
- TelPow prefix
  - Power supply level functions
- TelCfg prefix
  - Phone configuration (SMS configuration)

Telephone APIs (2 of 4)

- TelSpc prefix
  - Speech call handling and DTMF tones
- TelPhb prefix
  - Complete phonebook management

Telephone APIs (3 of 4)

- TelSty prefix
  - Phone and SIM security related (PIN code management)
- TelSms prefix
  - SMS management
- TelSnd prefix
  - Playing key tones and sound mute

Telephone APIs (4 of 4)

- TelInf prefix
  - Information about the phone (brand, model number and firmware version)
- TelEmc prefix
  - Emergency call handling
- TelOem prefix
  - Enable licensees to extend the Telephony Manager

Telephony Main Principles

- Load/Unload
  - The TelMgr is a standard Palm™ shared library
    - Loaded using **`SysLibFind, SysLibInstall`**
- Start/Stop
  - Once loaded, apps call **`TelOpen`** to open the library
    - First open => TelMgr starts the Telephony Task
    - Subsequent opens => increment a reference counter
  - Apps call **`TelClose`** when done
    - **`TelClose`** decrements the reference counter
    - If it reaches zero the Telephony Task is stopped

Timeouts

- The Telephony Manager does not handle timeouts
- Synchronous functions block until completion or error
  - Driver provides timeout for associated phone
- Asynchronous functions return immediately and a **`TelEventType`** is sent on function completion or error
  - Asynchronous functions can be cancelled
  - Application should set timer, and cancel function if the timer expires

Telephony Conclusion

- Create great apps for phones!
  - Let us worry about providing a common interface
- Use your Palm™ handheld to create a better phone
  - Bigger screen
  - Better input method

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SMS (Short Messaging Service)

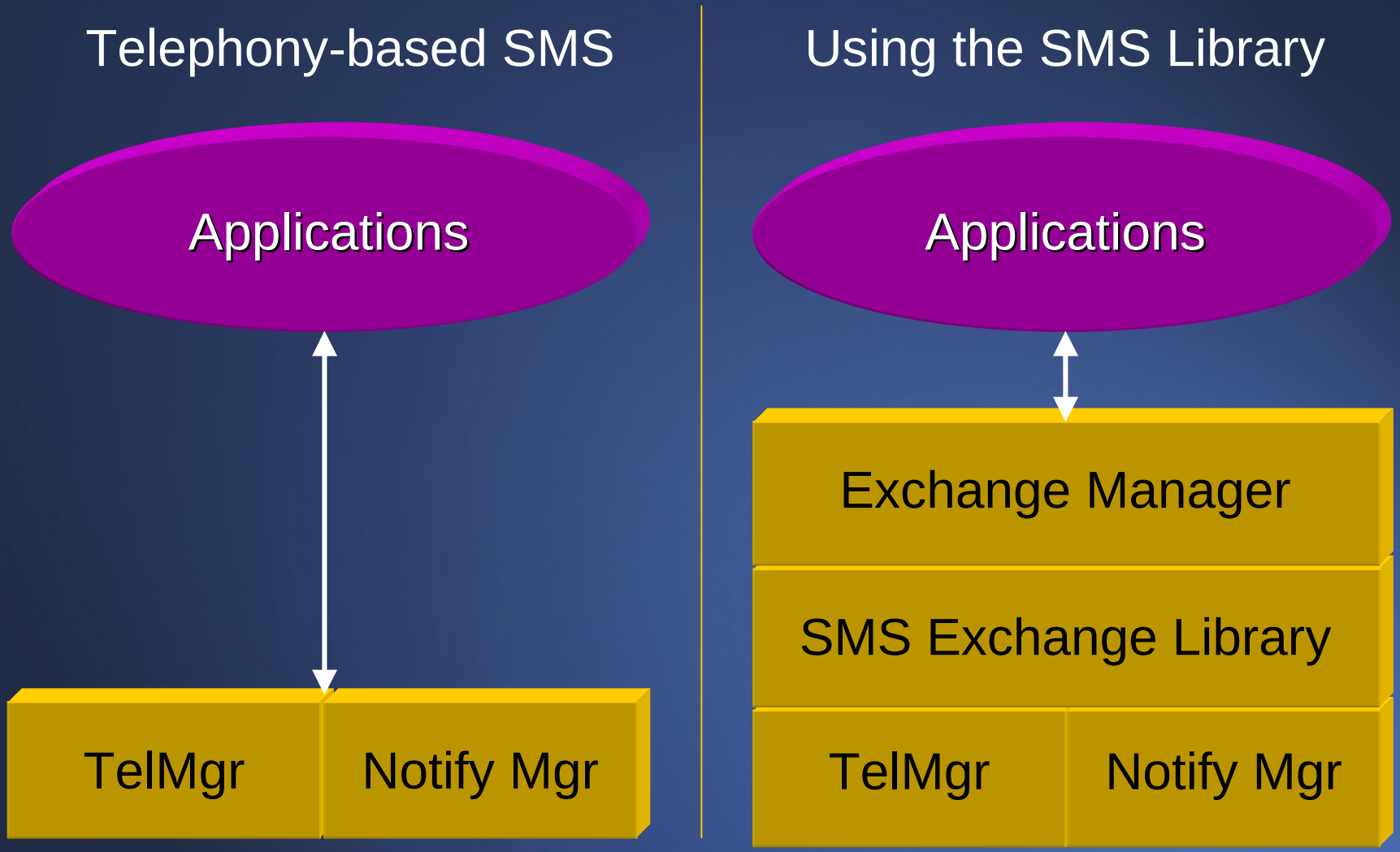
SMS is the standard two-way messaging format in phone networks. It offers a long-distance equivalent to beaming data between two Palm Powered™ devices.

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SMS Overview

- There are 2 ways to send and receive SMS messages
  - Telephony based SMS
  - SMS Exchange Library

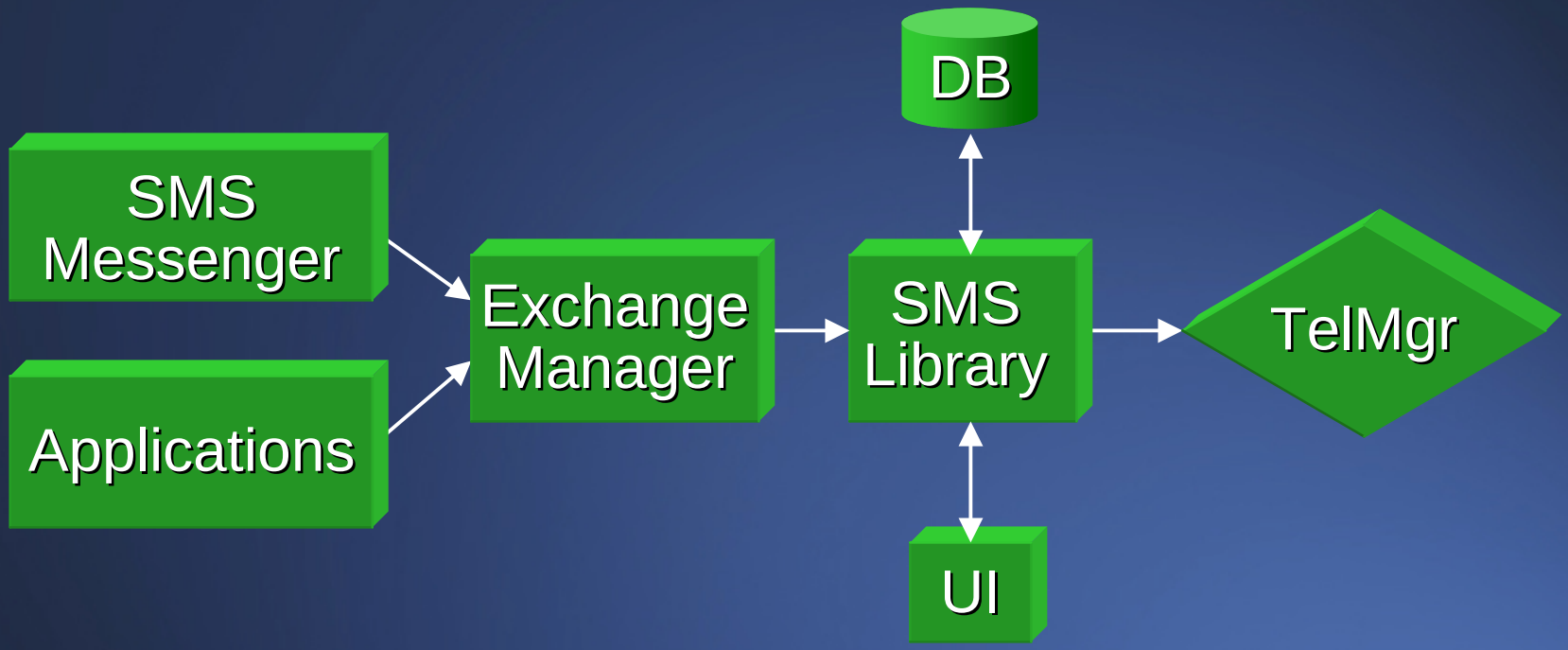
SMS Overview



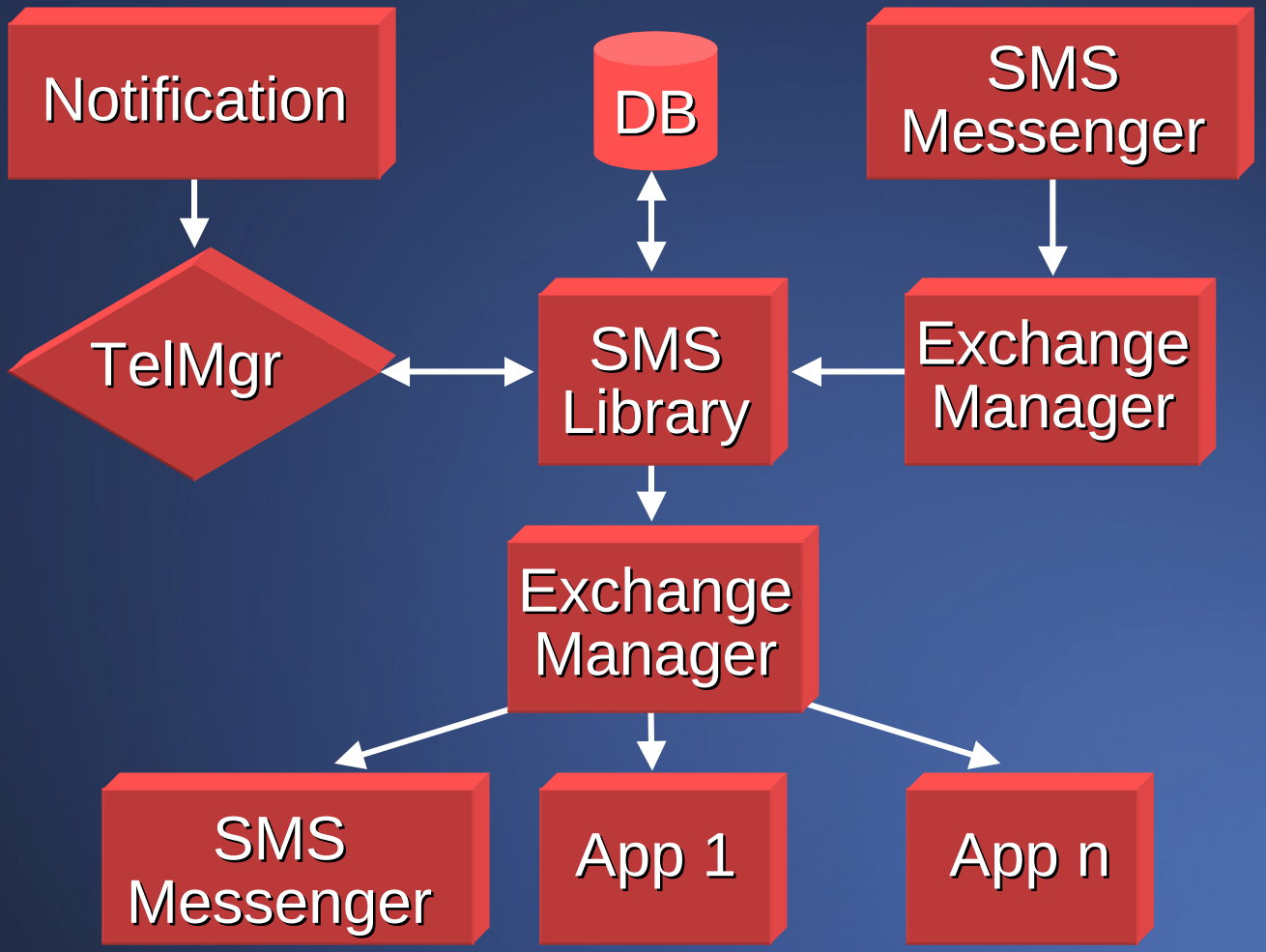
Telephony-based SMS

- Supported features:
  - SMS send
  - SMS receive message
  - SMS receive reports
  - SMS acknowledgment
- SMS functions are common to every telephony standard (GSM, CDMA, TDMA)
  - Most networks and phones support “mobile terminated”
  - Many support “mobile originated”

SMS Send Overview



SMS Retrieval Overview



Application

Independent of transport

- Everything as usual
  - Only uses the Exchange Manager
  - Registers its own data type
  - Used by PIM apps, like Address Book
- Data is received into SMS Messenger
  - Forwarded to Exchange Manager

Application

SMS Aware

- Registers its own data type
- Uses the SMS Library parameter in the Exchange Manager (socketref field)
- Specialized SMS applications
  - Chat over SMS
  - Stock tracker
  - …

SMS Messenger

- Only one should exist on a device
- Manages all SMS transactions
- Register to be notified on incoming SMS
- Register for data types:
  - “sms” (Any messages)
  - “rps” (SMS reports)

- Settings can be overwritten by third parties!

SMS Exchange Library

- Hides complexity of data transport from apps and the Exchange Manager
  - Simplifies 3rd-party application development
  - Expands functionality of wireless messaging
  - Provides powerful API for handling SMS
- Exchange Manager has been rewritten
  - SMS is just another transport!

SMS Library Features

- Implemented as a shared library
- Provides “beaming” over SMS
- Send SMS from any application
- Receive SMS from within any application

SMS Library Components

- SMS Engine
- SMS Converter
- SMS Lib
- SMS Database

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SMS Engine

- Manages and links all library components
- Handles message fragmentation and reassembly
- Single message size:
  - Text message = 160 bytes
  - 8-bit message = 142 bytes
- Largest multi-part message is 255 fragments
  - Text message = 37Kb
  - 8-bit message = 33Kb

SMS Converter

- Responsible for message encoding and decoding
  - Narrow Band Sockets (NBS)
    - Used to ensure compatibility and data exchange with major smart messaging devices on the market
    - Guaranties other protocols based on NBS (like WAP and WAP push) will be supported with code change
  - Palm Encoding Scheme
    - Wraps data sent by Exchange Manager

SMS Lib

- Entry point for applications and the exchange manager
- Provides abstraction

SMS Database

- Stores parts of a multi-part message until entire message can be reassembled
- API to retrieve incomplete messages

SMS Message Dispatching

- Incoming messages are dispatched to application via the Exchange Manager
- Dispatch could be “direct” or “via SMS Messenger”
  - Application decides
  - PIM apps use “via SMS Messenger”

SMS Conclusion

- Use SMS telephony APIs to read and write SMS messages
- Use SMS Exchange Library to send and receive SMS data from your application
- Use Exchange Manager to send and receive data
  - User decides on transport (IR, SMS, Bluetooth…)
  - No changes to your application

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SMS and Telephony Resources

- Provider Pavilion
  - ROM image that includes telephony and SMS libraries
  - API documentation
  - Montpellier!!!
    - Sample code???
    - SMS Messenger source code???

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